

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	NARRAVULA NAVYA SRI	Reg. No.:	192324220
Section / Batch:	AI & DS	Date:	20-07-26

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1. Master Theorem Application for Mixed Polynomial Term

Consider the recurrence relation $T(n) = 2T(n/2) + n^2 \log n$. Analyze the problem using the Master Theorem and determine the time complexity. Clearly identify parameters and compare $f(n)$ with $n^{\log_b a}$. Justify the correct case selection and explain reasoning. Use recursion tree diagrams if needed and interpret the result clearly.

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Master Theorem Application for Mixed Polynomial Term

NARRAVULA NAVYA SRI

192324220

Artificial Intelligence and Data Science

CSA0614-Design and Analysis of Algorithm

Problem Statement

Objective:

Determine the asymptotic time complexity of the recurrence:

$$T(n) = 2T\left(\frac{n}{2}\right) + n^2 \log n$$

using the **Master Theorem**.

Tasks

- Identify parameters a , b , and $f(n)$.
- Compute $n^{\log_b a}$.
- Compare $f(n)$ with $n^{\log_b a}$.
- Select the appropriate Master Theorem case.
- Conclude the time complexity.

Algorithm Overview

The Master Theorem solves recurrences of the form

$$T(n) = aT\left(\frac{n}{b}\right) + f(n)$$

Step 1

Identify:

$$a = 2$$

$$b = 2$$

$$f(n) = n^2 \log n$$

Step 2

Compute

$$n^{\log_b a} = n^{\log_2 2} = n$$

Step 3

Compare

$$f(n) = n^2 \log n$$

with

$$n^{\log_b a} = n$$

Since

$$n^2 \log n = \Theta(n^{1+\epsilon})$$

for $\epsilon = 1$, $f(n)$ grows polynomially faster.

Algorithm / Pseudocode

Master Theorem($T(n)$)

Input:

$$T(n) = 2T(n/2) + n^2 \log n$$

1. $a \leftarrow 2$

2. $b \leftarrow 2$

3. $f(n) \leftarrow n^2 \log n$

4. Compute $n^{\log_b a}$

5. If $f(n) = \Theta(n^{\log_b a})$

Case 2

6. Else if $f(n) = \Theta(n^{\log_b a - \epsilon})$

Case 1

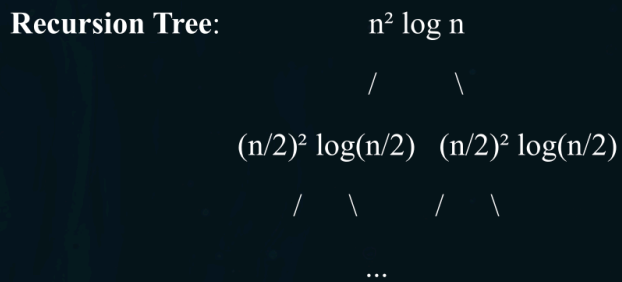
7. Else if $f(n) = \Theta(n^{\log_b a + \epsilon})$

Check Regularity Condition

Case 3

8. Output $\Theta(f(n))$

Working Example



Cost at each level:

Level 0:
 $n^2 \log n$

Working Example

Level 1

$$2 \left(\frac{n}{2}\right)^2 \log\left(\frac{n}{2}\right) = \frac{n^2}{2} (\log n - 1)$$

Level 2

$$4 \left(\frac{n}{4}\right)^2 \log\left(\frac{n}{4}\right) = \frac{n^2}{4} (\log n - 2)$$

General level i

$$\frac{n^2}{2^i} (\log n - i)$$

The cost decreases geometrically.

Total work:

$$n^2 \log n + \frac{n^2}{2} \log n + \frac{n^2}{4} \log n + \dots = \Theta(n^2 \log n)$$

Complexity Analysis

Parameters:

Parameter	Value
(a)	2
(b)	2
(f(n))	(n ² log n)

Compute

$$n^{\log_b a} = n$$

Comparison

$$n^2 \log n > n^{1+\epsilon}$$

Choose

$$\epsilon = 1$$

Regularity Condition

Check

$$af(n/b) \leq cf(n)$$
$$2 \left(\frac{n}{2}\right)^2 \log \left(\frac{n}{2}\right) = \frac{n^2}{2} (\log n - 1)$$

Complexity Analysis

For sufficiently large n ,

$$\frac{n^2}{2} (\log n - 1) \leq c(n^2 \log n)$$

for some constant $c < 1$ (e.g., $c = \frac{3}{4}$).

Hence the regularity condition holds.

Master Theorem Case

Case 3

Therefore,

$$T(n) = \Theta(n^2 \log n)$$

Comparative Analysis

Case	Condition	Complexity
Case 1	$f(n)$ smaller	$\Theta(n^{\log_{ba}})$
Case 2	Same order	$\Theta(n^{\log_{ba}} \log^{k+1} n)$
Case 3	$f(n)$ polynomially larger	$\Theta(f(n))$

For this recurrence:

$$n^2 \log n \gg n$$

Hence **Case 3** applies.

Applications

Master Theorem is widely used in analysing:

- Divide and Conquer algorithms
- Merge Sort
- Binary Search
- Strassen Matrix Multiplication
- Closest Pair of Points
- Karatsuba Multiplication
- FFT Algorithms
- Computational Geometry algorithms

Results & Discussion

Result

Given

$$T(n) = 2T(n/2) + n^2 \log n$$

we obtain

$$\begin{aligned}a &= 2 \\b &= 2 \\n^{\log_b a} &= n \\f(n) &= n^2 \log n\end{aligned}$$

Since

$$f(n) = \Omega(n^{1+\epsilon})$$

and the regularity condition is satisfied,

Master Theorem Case 3 applies.

Final Complexity

$$\boxed{\Theta(n^2 \log n)}$$

The recursion tree also confirms that the work is dominated by the root level, with subsequent levels decreasing geometrically.

Conclusion & References

Conclusion:

Identified:

$$\begin{aligned}a &= 2 \\b &= 2 \\f(n) &= n^2 \log n\end{aligned}$$

Computed:

$$n^{\log_b a} = n$$

Compared:

$n^2 \log n$ grows polynomially faster than n

Applied **Master Theorem Case 3**

Final result:

$$\boxed{T(n) = \Theta(n^2 \log n)}$$

References:

- Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein, **Introduction to Algorithms (CLRS)**, 4th Edition.
- Anany Levitin, **Introduction to the Design & Analysis of Algorithms**, 4th Edition.
- Michael T. Goodrich, Roberto Tamassia, **Algorithm Design and Applications**.

THANK YOU

